

**NANYANG TECHNOLOGICAL UNIVERSITY
HE2003 - ECONOMETRICS I MIDTERM QUIZ**

Time Allowed: 1 1/2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer all the questions.
3. This is a **closed-book** quiz.
4. The number of marks allocated is shown at the end of each question.

MATRICULATION NUMBER:

1. Explain the following terms; Use specific examples, formulas, or graphs if necessary.
 - (a) Least squares assumptions. (8 Marks)

(b) Linear regression model with common slope and multiple intercepts. (8 Marks)

(c) The definition of Type I and Type II errors. The trade-off between Type I and Type II errors. (8 Marks)

(d) The meaning of unbiasedness and efficiency. The Gauss-Markov Theorem. (8 Marks)

(e) Homoskedasticity and heteroskedasticity of error terms. (8 Marks)

(TOTAL: 40 Marks)

2. Suppose that for the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, the binary regressor X_i takes the values of 1 and 0 with the following probability: $P(X_i = 1) = 0.5$ and $P(X_i = 0) = 0.5$. In addition, when $X_i = 1$, the error term u_i follows the distribution $N(0, 3)$ (that is, $u_i \sim N(0, 3)$ if $X_i = 1$), and when $X_i = 0$, u_i follows the distribution $N(0, 2)$ ($u_i \sim N(0, 2)$ if $X_i = 0$).
- (a) Suppose that Y_i denotes person i 's wage, u_i denotes i 's innate ability, and $X_i = 1$ if i graduated from University A while $X_i = 0$ if i graduated from University B. Are the distributions of u_i different for the graduates of the two universities? Explain. (5 Marks)
- (b) Show that the OLS Assumption 1 holds in this case. Show the interpretation of the parameters β_0 and β_1 by using conditional expectation. (10 Marks)

- (c) Suppose we estimate the coefficients β_0 and β_1 by using the OLS and we obtain the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$. In addition, suppose we have constructed alternative estimators for the coefficients as $\tilde{\beta}_0 = 0.5 \cdot \hat{\beta}_0$ and $\tilde{\beta}_1 = 2 \cdot \hat{\beta}_1$. Compare $(\hat{\beta}_0, \hat{\beta}_1)$ versus $(\tilde{\beta}_0, \tilde{\beta}_1)$ in terms of unbiasedness and efficiency. In addition, illustrate the difference of their distributions using graph. (5 Marks)

(TOTAL: 20 Marks)

3. Consider the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$.

- (a) Show the following holds:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2},$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. (10 Marks)

- (b) Show the average of the OLS predicted values $\hat{Y}_i$ is equal to the sample average of Y ($\bar{Y}$). Show the average of the residuals $\hat{u}_i$ is equal to zero. (10 Marks)

(TOTAL: 20 Marks)

4. Suppose that a random sample of size $n = 100$ is drawn from certain population and the observations $(Y_i, X_i), i = 1, \dots, 100$ satisfy the OLS assumptions. The result of the OLS regression is:

$$\hat{Y}_i = 3.2 + 4.2X_i, i = 1, \dots, 100,$$

with the standard error of $\hat{\beta}_0$ equal to 1 and the standard error of $\hat{\beta}_1$ equal to 2.

- (a) Test the slope of the regression using $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ at the 5% significance level. Similarly, test the intercept of the regression at the 5% significance level. (5 Marks)

(b) Does the 95% confidence interval for β_1 contain 0? Explain. Is the p -value of the t -test for $H_0 : \beta_0 = 0$ vs $H_1 : \beta_0 \neq 0$ smaller than 5%? Explain. (5 Marks)

(c) Write down the formula of the t -test statistic for testing $H_0 : \beta_1 = 2$ vs $H_1 : \beta_1 \neq 2$. Explain how we can show that the (large-sample) distribution of the t -test statistic follows $N(0, 1)$ when the null hypothesis H_0 is true. (10 Marks)

(TOTAL: 20 Marks)

END OF PAPER